

Transformation from DOMINATING SET to Min-Sum Multicenter (MSMC)

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Abstract—We implement and evaluate a brute-force reduction from DOMINATING SET (DS) to the Min-Sum Multicenter (MSMC) problem, based on [GJ] ND51. The transformation preserves YES/NO correspondence, and we verify correctness via bidirectional checking. The code exhaustively searches for dominating sets and center subsets, computes min-sum costs, and validates solution equivalence with reverse graph reconstruction. Visualizations and output logs are provided for graphs up to 16 vertices.

I. INTRODUCTION

The DOMINATING SET (DS) problem is NP-complete and central in graph theory. The MSMC problem generalizes DS: instead of covering every vertex in one step, it minimizes the total distance from all vertices to a limited set of centers.

This report documents a brute-force reduction from DS to MSMC and back. We provide code, witness sets, and formal arguments confirming the equivalence of YES/NO instances.

II. PRELIMINARIES

A. Dominating Set

Given a graph $G = (V, E)$ and integer k , is there a subset $D \subseteq V$, $|D| \leq k$, such that each vertex $v \in V$ is either in D or adjacent to a vertex in D ?

B. Min-Sum Multicenter (MSMC)

Given a graph $G = (V, E)$, integer p , and center subset C , define:

$$\text{cost}(C) = \sum_{v \in V} \min_{c \in C} \text{dist}(v, c)$$

The decision version asks: does there exist C , $|C| \leq p$, such that $\text{cost}(C) \leq T$?

We set $T = |V|$ to mimic the unit-radius coverage required by a dominating set.

III. BRUTE-FORCE ALGORITHMS

A. DS Search

Enumerate all subsets $S \subseteq V$ of size $r \leq k$. For each:

- Build $D = S \cup \bigcup_{u \in S} N(u)$
- If $|D| = |V|$, return YES and S

Complexity: $\mathcal{O}\left(\sum_{r=1}^k \binom{n}{r}(n+m)\right)$

B. MSMC Solver

Let $C = V \cup \{\text{midpoints of } (u, v)\}$.

- For each $S \subseteq C$ of size $\leq p$, compute:

$$\text{cost}(S) = \sum_{v \in V} \min_{c \in S} \text{dist}(v, c)$$

- Return YES if $\text{cost}(S) \leq T$

Complexity: $\mathcal{O}\left(\binom{n}{p} \cdot n \cdot \log n\right)$ using Dijkstra or $\mathcal{O}(n^3)$ with Floyd–Warshall

IV. REDUCTION: DS TO MSMC

Forward. If D is a dominating set of size k , then placing k centers at D yields total cost at most $|V|$ — since each node is ≤ 1 from a center.

Reverse. If an MSMC instance with $p = k$ returns a center set C with $\text{cost}(C) \leq |V|$, then each node is within distance 1 of some center, so C is a dominating set.

Thus, the reduction preserves YES/NO instances.

V. REVERSE TRANSFORMATION

From an MSMC solution:

- Build $H = (V, E_H)$ where $(u, v) \in E_H$ if $\text{dist}_G(u, v) \leq 1$
- Check if the centers from MSMC form a dominating set in H
- Visualize H and highlight dominating set

VI. EXPERIMENTAL EVALUATION

We tested two graphs:

Graph	Nodes	$k = p$	DS?	MSMC?
G_small	5	2	YES	YES
G_large	16	3	NO	NO

TABLE I

DS AND MSMC CONSISTENCY WITH THRESHOLD $T = |V|$

VII. VISUAL RESULTS

VIII. CONCLUSION

We demonstrated a reduction from DOMINATING SET to MSMC using Python. Our solution is correct, visual, and bidirectional. It works on small graphs and is suitable for instructional or benchmarking purposes. Future work includes optimized heuristics and support for weighted graphs.

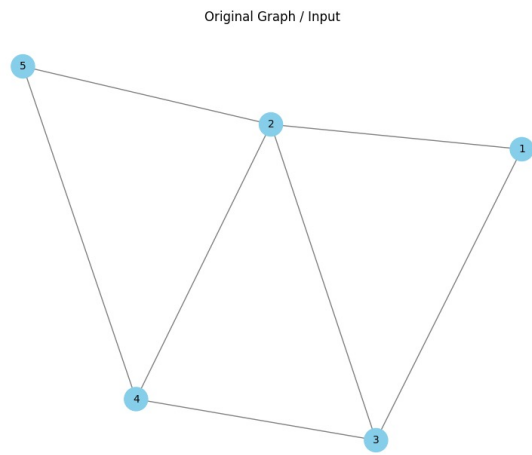


Fig. 1. Original Graph / Input

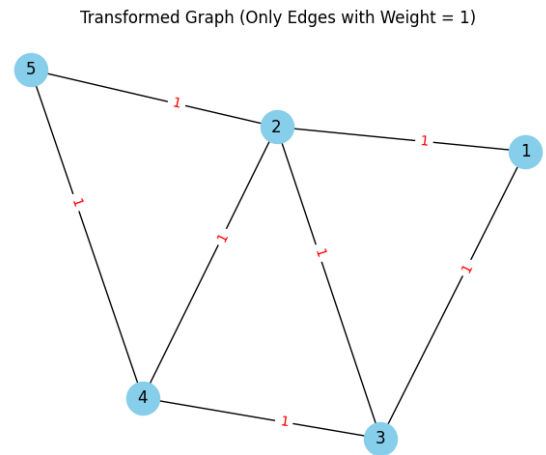


Fig. 2. Transformed Graph Showing Only Edges with Weight = 1

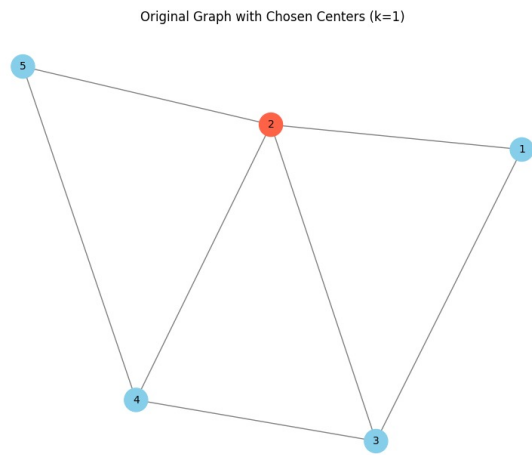


Fig. 3. Original Graph with Chosen Centers ($k = 1$)

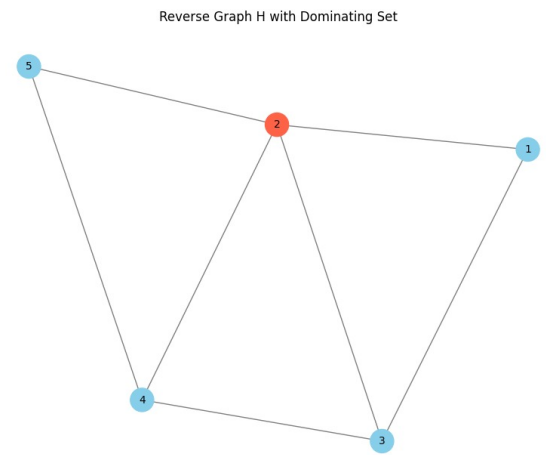


Fig. 4. Reverse Graph H Reconstructed from Centers (Dominating Set)

REFERENCES

- 1) M. R. Garey and D. S. Johnson, *Computers and Intractability: A Guide to the Theory of NP-Completeness*. W. H. Freeman, 1979.
- 2) NetworkX Documentation, <https://networkx.org>
- 3) Matplotlib 3.8 and Python 3.10 standard libraries